

Logistics Information Systems, Reverse Logistics, Network Design & 3PL / 4PL

Designing Logistics Networks • Selecting Logistics Providers & Intermediaries

What We Will Cover — 5 Modules

01

Logistics Information Systems (LIS)

Components, technologies, ERP/WMS/TMS, examples

02

Reverse Logistics

Drivers, process flow, types of flows, real cases

03

Designing the Logistics Network

Facility location, network configurations, trade-offs

04

3PL & 4PL

Evolution of outsourcing, roles, comparison, examples

05

Provider Selection & Intermediaries

Types, criteria, RFP process, decision framework

By the End of This Module, You Will Be Able To

- 1 Explain the role and components of a Logistics Information System (LIS) in coordinating supply chain flows
- 2 Describe the reverse logistics process and its strategic drivers — returns, recalls, recycling, sustainability
- 3 Apply key principles of logistics network design, including facility location and network configuration
- 4 Differentiate between 3PL and 4PL models and identify when each is appropriate
- 5 Evaluate and select logistics providers and intermediaries using a structured decision framework

MODULE 1

Logistics Information Systems (LIS)

The digital backbone connecting demand, inventory, transport and warehousing decisions

What Is a Logistics Information System?

- **Definition:** An integrated system of people, processes and technology that captures, stores, and communicates data needed to plan, execute, and control logistics activities.
- **Purpose:** Converts fragmented data (orders, inventory, shipments) into timely, accurate information for decision-making.
- **Scope:** Spans order processing, inventory visibility, warehouse operations, transportation, and customer service.
- **Strategic role:** Enables the shift from reactive firefighting to proactive, data-driven supply chain management.
- **Outcome:** Higher order accuracy, lower inventory cost, faster cycle times, and real-time end-to-end visibility.

Core Components of a Logistics Information System

OM

Order Management

Captures customer orders, checks stock availability, and triggers fulfillment workflows.

IM

Inventory Management

Tracks stock levels, locations, and movements in real time across nodes.

WM

Warehouse Management

Directs putaway, picking, packing and slotting decisions inside a facility.

TM

Transportation Management

Plans routes, selects carriers, tenders loads, and tracks in-transit shipments.

CS

Customer Service Interface

Provides order status, ETAs, and exception alerts to customers and sales teams.

MIS

Reporting & Analytics (MIS)

Converts operational data into KPIs — OTIF, fill rate, cost-to-serve — for managers.

Important slide

Enabling Technologies Behind LIS

- **EDI (Electronic Data Interchange):** Structured, automated exchange of purchase orders, invoices, and ASNs between trading partners.
- **Barcoding & RFID:** Automatic identification of items, pallets and containers for real-time inventory accuracy.
- **GPS / GIS & Telematics:** Live vehicle location, route tracking, and geofencing for transportation visibility.
- **IoT sensors:** Temperature, humidity and shock monitoring for cold-chain and fragile-cargo shipments.
- **Cloud & APIs:** Enable ERP, WMS and TMS systems from different vendors to exchange data seamlessly.

Common LIS Software Categories

ERP

ERP (e.g., SAP, Oracle)

Integrates finance, procurement, and logistics data across the enterprise.

WMS

Warehouse Management System

Optimizes storage, picking paths, and labour inside distribution centres.

TMS

Transportation Management System

Plans loads, optimizes routes, and manages freight settlement.

YMS

Yard Management System

Schedules dock doors and tracks trailers within a yard or DC campus.

APS

Advanced Planning System

Forecasts demand and synchronizes supply plans across the network.

VMS

Visibility / Control-Tower Platform

Aggregates data from multiple systems into one real-time dashboard.

LIS in Action — Real Company Examples

Amazon

- Proprietary WMS coordinates millions of SKUs across fulfillment centres with robotics integration
- Real-time inventory sync between warehouse, seller, and customer-facing app
- Predictive analytics pre-positions inventory near expected demand

Maersk

- TradeLens-style digital platform gave shippers real-time container visibility
- IoT-enabled reefer containers monitor temperature across ocean transit
- API integration links customs, ports, and inland transport data

DHL

- MySupplyChain control tower dashboard for multi-modal shipment tracking
- RFID-enabled parcel sortation raises scan accuracy and throughput
- Predictive ETA engine uses traffic, weather, and historical transit data

Benefits and Implementation Challenges

- **Benefit — Visibility:** Single source of truth for inventory and shipment status across the network.
- **Benefit — Decision speed:** Real-time dashboards replace end-of-day or weekly manual reports.
- **Benefit — Cost control:** Better route and load planning reduces empty miles and warehouse labour cost.
- **Challenge — Integration:** Legacy systems and disparate data formats slow down system-to-system integration.
- **Challenge — Data quality & cybersecurity:** Poor master data undermines trust in the system; connected devices widen the attack surface.

Video: Understanding Logistics Information Systems

Play before class discussion — how does each video's example map to the LIS components covered?



What is Logistics Management Explained

<https://www.youtube.com/watch?v=RrZfNbDIUjs>



SAP Logistics Management Explained: What It Is and Where It Fits

<https://www.youtube.com/watch?v=oZ4ViVjBpTk>



An Introduction to Logistics and Supply Chain Management

<https://www.youtube.com/watch?v=GCK18JyPVXI>

Tip for classroom use: play the 3–6 minute clip, then debrief with the discussion prompts on the following slide.

MODULE 2

Reverse Logistics

Managing the backward flow of goods — from returns to recovery and disposal

What Is Reverse Logistics?

- **Definition:** The process of planning, implementing and controlling the efficient flow of goods, information and returns from the point of consumption back to the point of origin.
- **Also called:** Aftermarket logistics, returns management, or closed-loop supply chain management.
- **Core idea:** Recaptures value from products that would otherwise be scrapped — through resale, repair, remanufacture, or recycling.
- **Growing importance:** E-commerce return rates and sustainability regulation have made reverse logistics a board-level priority, not an afterthought.

Forward vs. Reverse Logistics — Key Differences

Dimension	Forward Logistics	Reverse Logistics
Flow direction	Producer → Customer	Customer → Producer / recovery point
Quality of goods	Uniform, standardized product	Variable — damaged, used, or defective
Forecasting	Relatively predictable demand	Volume and timing are uncertain
Packaging	Standard, often uniform	Often damaged or non-standard
Destination options	Single, known destination	Multiple options — resale, repair, recycle, dispose
Cost visibility	Well understood, budgeted	Often hidden / under-measured

Key Drivers of Reverse Logistics

1

Product Returns

Customer returns from e-commerce, wrong-size, or damaged-in-transit items.

2

Recalls & Quality Issues

Regulatory or safety-driven withdrawal of defective products from the market.

3

End-of-Life Recovery

Recycling and disposal at the end of a product's useful life (e-waste, batteries).

4

Sustainability & Circular Economy

Corporate ESG goals push firms toward remanufacturing and material recovery.

5

Regulatory Compliance

Extended Producer Responsibility (EPR) laws mandate take-back programmes.

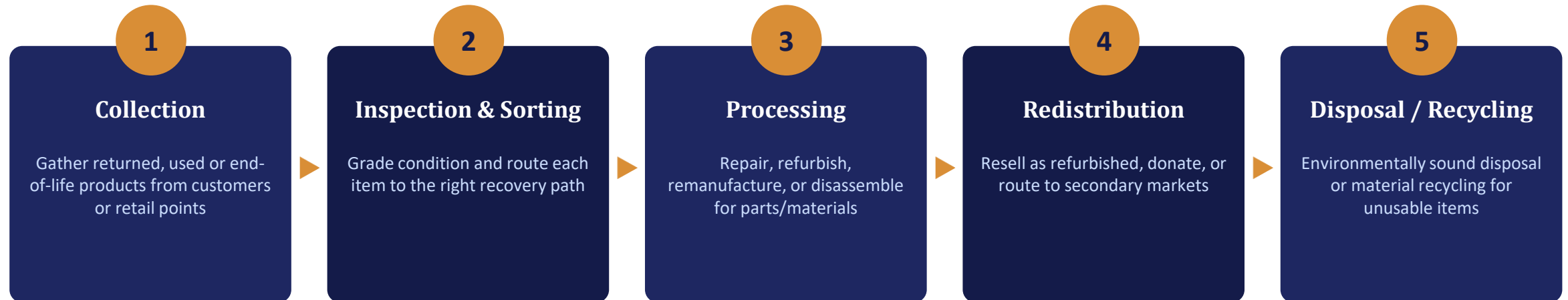
6

Warranty & Repair

Products routed back for servicing, refurbishment, and re-issue.

The Reverse Logistics Process Flow

Reverse logistics is typically modelled as a five-stage flow, moving a returned item from the customer back through the network to its best next use.



Types of Reverse Flows

R

Returns Management

Handling customer returns for refund, exchange, or store credit.

S

Repair & Refurbishment

Restoring products to a sellable or usable condition.

M

Remanufacturing

Disassembling and rebuilding products to as-new specification.

C

Recycling

Recovering raw materials (metal, plastic, e-waste components) for reuse.

D

Disposal

Environmentally compliant destruction of unusable or hazardous items.

P

Packaging Recovery

Reusable pallets, crates and containers cycled back to the source.

Reverse Logistics in Practice

Read this slide

Amazon

- Automated returns via QR code drop-off at partner stores (no box/label needed)
- AI-based grading routes items to resale, liquidation, or donation
- "FBA Grade and Resell" recovers value from customer-returned inventory

Apple

- Trade-In and Apple Renew programmes recover devices for refurbishment
- Daisy robot disassembles iPhones to recover rare-earth materials
- Closed-loop goal: use only recycled/renewable materials in products

H&M

- In-store Garment Collecting take-back scheme for used clothing
- Textiles sorted for rewear, reuse, or fibre-to-fibre recycling
- Extended Producer Responsibility aligned with EU textile regulation

Video: Reverse Logistics Explained

After viewing, ask: which stage of the 5-step reverse flow does each video emphasize?



What Is Reverse Logistics? Global Supply Chain Solutions Explained

<https://www.youtube.com/shorts/elvNcsHIG64>



Reverse Logistics Explained

<https://www.youtube.com/watch?v=wQ3Z60YFBaY>



What Is REVERSE LOGISTICS? Part 1 (Supply Chain Basics)

https://www.youtube.com/watch?v=42t_t0-sxs0

Tip for classroom use: play the 3–6 minute clip, then debrief with the discussion prompts on the following slide.

MODULE 3

Designing the Logistics Network

Where to locate facilities and how to connect them for cost, speed and service

What Is Logistics Network Design?

- **Definition:** The strategic process of deciding the number, location, and role of facilities — plants, warehouses, DCs — and the flows that connect them to customers.
- **Planning horizon:** A long-term, capital-intensive decision, typically revisited every 3-5 years or after major demand shifts.
- **Objective:** Minimize total logistics cost (facility + inventory + transportation) while meeting target service levels.
- **Why it matters:** A poorly designed network locks in high cost-to-serve and slow delivery for years — hard and expensive to undo.

Key Strategic Decisions in Network Design

1

Number of Facilities

How many plants/DCs/warehouses are needed to balance cost and coverage?

2

Facility Location

Where should each facility sit relative to suppliers, customers and infrastructure?

3

Capacity Allocation

How much volume and which products/markets does each facility serve?

4

Inventory Positioning

Where in the network should safety stock and buffer inventory sit?

5

Transportation Mode & Lanes

Which mode (road/rail/air/sea) and routing links each network node?

6

Make vs. Outsource

Should the firm own facilities/fleet or contract out to 3PL/4PL partners?

Factors That Shape Network Design Decisions

- **Customer service targets:** Delivery lead-time promises (same-day, next-day) dictate how close DCs must be to demand.
- **Cost trade-offs:** Fewer, larger DCs lower facility and inventory cost but raise outbound transportation cost — and vice versa.
- **Demand patterns:** Volume, seasonality, and geographic concentration of customers shape where nodes are needed.
- **Infrastructure & regulation:** Port/road/rail access, free-trade zones, taxation, and labour availability affect site economics.
- **Risk & resilience:** Diversifying facility locations reduces exposure to a single point of disruption (natural disaster, geopolitical risk).

Facility Location Analysis Methods

- **Center-of-Gravity method:** Locates a facility at the demand- and volume-weighted geographic centre of its customer base to minimize transport distance.
- **Weighted Factor Rating:** Scores candidate sites against criteria (labour cost, infrastructure, proximity, incentives) each weighted by importance.
- **Greenfield / Network Optimization Software:** Simulates thousands of network configurations (e.g., anyLogistix) to find the lowest-cost feasible design.
- **Break-even / Total Cost Analysis:** Compares the total landed cost of alternative site options across a range of volumes.

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Common Network Configurations

Configuration	How It Works	Best Suited For
Centralized (single DC)	One large national/regional DC serves all customers	Low-volume, high-value SKUs; slower service acceptable
Decentralized (multi-DC)	Several regional DCs positioned near demand clusters	High service-level needs, bulky or fast-moving goods
Hub-and-Spoke	Central hub consolidates volume, spokes handle last-mile	Parcel and less-than-truckload (LTL) networks
Cross-Docking	Goods flow through a facility with minimal storage time	High-velocity retail replenishment
Direct-to-Customer (DTC)	Ships directly from plant/3PL fulfillment centre to end user	E-commerce, drop-shipping models

Network Design in Practice

Flipkart / Amazon India

- Multi-tier network: national FCs, regional sortation hubs, local delivery stations
- Hyperlocal "dark stores" added in metros for 1-2 hour delivery
- Network re-optimized each festive season for demand spikes

Zara (Inditex)

- Single, highly automated distribution centre in Spain serves stores worldwide
- Centralized model traded off for speed of fashion-cycle replenishment
- Air freight bridges the distance gap to distant store networks

Coca-Cola India

- Hub-and-spoke bottling and distribution network across regional plants
- Redistribution stockists (RDS) act as the last-mile spoke to retailers
- Network density tuned to local demand and road infrastructure

Video: Logistics Network & Distribution Design

Discuss: what trade-off between cost and service does the video's example illustrate?



3PL vs 4PL: Layers of Logistics Explained

<https://www.youtube.com/watch?v=9J2pTNTksno>



Logistics Management in 12 Minutes

<https://www.youtube.com/watch?v=y9tyXL87l1A>

Tip for classroom use: play the 3–6 minute clip, then debrief with the discussion prompts on the following slide.

MODULE 4

3PL and 4PL

Outsourcing logistics execution and strategy to specialist partners

The Evolution of Logistics Outsourcing — 1PL to 5PL

- **1PL:** The shipper/manufacturer manages and moves its own goods with in-house resources.
- **2PL:** Asset-owning carriers (trucking, shipping, rail companies) transport goods for the shipper.
- **3PL:** A specialist provider manages warehousing, transportation and fulfillment on the client's behalf.
- **4PL:** A non-asset-owning integrator manages the entire supply chain, often coordinating multiple 3PLs.
- **5PL:** Aggregates and manages multiple 4PLs / supply chain networks, typically for very large, complex enterprises.

Important slide

Third-Party Logistics (3PL)

- **Definition:** An external provider that executes one or more logistics functions — warehousing, transportation, fulfillment, returns — on behalf of a client.
- **Assets:** Typically owns or leases warehouses, trucks, and labour; some operate an asset-light model using partner carriers.
- **Relationship style:** Largely transactional/operational — priced by volume, lane, or storage used.
- **Typical services:** Warehousing, pick-pack-ship, freight negotiation, inventory management, reverse logistics.
- **Examples:** DHL Supply Chain, Blue Dart, Delhivery, XPO Logistics, FedEx Supply Chain.

Fourth-Party Logistics (4PL)

- **Definition:** A non-asset-based integrator that designs, manages and optimizes a client's entire supply chain — often coordinating several 3PLs.
- **Assets:** No warehouses or trucks of its own; value lies in strategy, technology, and vendor orchestration.
- **Relationship style:** Strategic, long-term partnership — the 4PL becomes the client's single point of accountability.
- **Typical services:** Supply chain design, control-tower visibility, carrier/3PL selection, continuous cost optimization.
- **Examples:** Accenture Supply Chain Services, DHL 4PL (Lead Logistics), Kuehne+Nagel Integrated Logistics.

Read this slide

3PL vs. 4PL — Side-by-Side Comparison

Dimension	3PL	4PL
Focus	Execution — moving and storing goods	Strategy — designing and optimizing the network
Assets owned	Warehouses, trucks, labour	None — technology and expertise only
Scope of control	One or a few logistics functions	End-to-end supply chain oversight
Client relationship	Transactional, lane/volume priced	Strategic, long-term partnership
Coordinates	Own operations	Multiple 3PLs and carriers on client's behalf
Best suited for	SMEs, single-function outsourcing needs	Large, complex, multi-region supply chains

Video: 3PL vs 4PL Explained

Discuss: using the video, map a real company you know to either the 3PL or 4PL model.



3PL vs 4PL Logistics: The Layers of Logistics Explained

<https://www.youtube.com/watch?v=9J2pTNTksno>

Tip for classroom use: play the 3–6 minute clip, then debrief with the discussion prompts on the following slide.

MODULE 5

Selecting Logistics Providers & Intermediaries

Choosing the right partners to execute the network design

Types of Logistics Intermediaries

FF

Freight Forwarders

Arrange international shipment, consolidation, and documentation on the shipper's behalf.

CB

Customs Brokers

Manage customs clearance, duties, and regulatory compliance at borders.

CR

Carriers

Own and operate the physical transport — trucking, rail, ocean, or air fleets.

WH

Warehousing & Fulfillment Companies

Provide storage, pick-pack-ship, and value-added services.

AG

Logistics Agents / Brokers

Match shipper capacity needs with available carrier capacity in the spot market.

IT

Technology / Visibility Providers

Supply tracking, TMS, or control-tower platforms that connect the other players.

Role of Intermediaries in the Supply Chain

- **Aggregation & scale:** Consolidate small shipments from many shippers to access better freight rates.
- **Specialized expertise:** Navigate complex customs, regulatory, or mode-specific requirements shippers can't justify building in-house.
- **Risk transfer:** Absorb operational and, in some contracts, financial risk of delay or damage.
- **Market access:** Provide reach into regions or networks the shipper does not operate in directly.
- **Flexibility:** Let firms scale logistics capacity up or down without fixed asset investment.

Criteria for Selecting a Logistics Provider

- **Cost competitiveness:** Transparent, benchmarked pricing relative to service level offered.
- **Service reliability:** Track record on On-Time-In-Full (OTIF) delivery and damage/error rates.
- **Network reach & capacity:** Geographic coverage and ability to flex capacity during peak demand.
- **Technology & visibility:** Real-time tracking, system integration (API/EDI) with the client's LIS.
- **Financial stability:** Ability to sustain operations and invest, especially for long-term 4PL relationships.
- **Sustainability credentials:** Emissions reporting, fleet efficiency, and alignment with client ESG targets.

A Structured Provider Selection Process



Provider Selection — A Worked Example

- **Scenario:** A D2C apparel brand scaling from 500 to 20,000 orders/month needs to outsource fulfillment.
- **Shortlist:** Three 3PLs evaluated on weighted criteria: cost (30%), SLA reliability (25%), tech integration (20%), scalability (15%), location fit (10%).
- **Decision:** Provider B scored highest despite a 6% higher rate card — its API-ready WMS cut integration time from 6 weeks to 10 days.
- **Lesson:** Lowest cost is rarely the right criterion in isolation — total cost of ownership and service risk matter as much as the rate card.

Bringing It All Together

- LIS is the connective tissue — without accurate, real-time data, every downstream logistics decision suffers.
- Reverse logistics is a strategic value-recovery process, not just a returns-handling cost centre.
- Network design decisions are long-lived and capital-intensive — get facility location and configuration right early.
- 3PL buys execution; 4PL buys strategic orchestration — choose based on complexity, not just cost.
- Provider and intermediary selection should be a structured, criteria-weighted process — not a rate-card decision alone.

Discussion Question: For a company you know well, would you recommend a 3PL or a 4PL model — and why?